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A Review of Indigenous Perspectives in Animal Biosciences

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Keywords

Indigenous, animal sciences, American Indian, Native American, One Health, Traditional Ecological Knowledge

Abstract

This article addresses the underrepresentation of Indigenous perspectives in animal sciences by challenging the exclusive use of Western scientific paradigms in research and education. Because of the systematic exclusion of Indigenous peoples, Indigenous perspectives have rarely been represented through empirical study, leading us to believe this is a key reason for the underrepresentation of Native people in these fields. We conducted a literature review, searching for Indigenous contributions in animal sciences and finding a handful of articles in three areas: human–animal bonds, genetic testing and breeding programs, and Traditional Ecological Knowledge. Given the interconnected paradigm of Indigenous worldviews, we suggest that the ongoing siloes of scientific disciplines and the hierarchy of methodology contribute to the dearth of Indigenous perspectives. We suggest increased support for proper tribal consultation, contextualization of the history of research in Native communities, and the creation of scholarly spaces to support these conversations.

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INTRODUCTION

This article reviews the contribution of Indigenous perspectives in several fields of animal sciences. The authors are interested in the following questions: Do Indigenous scientific perspectives exist within various fields of animal sciences? If not, why? We consider animal sciences broadly, including research that has implications for animal husbandry, human–animal bonds, animal interventions, and animal genetics. We begin by explaining what Indigenous and Native perspectives in science are and how these perspectives might intersect with various fields in animal science. We cite a collection of books and articles from education, Native American studies, and environmental studies to trace the development of Native perspectives as they have been articulated in academic communities, with a focus on Native perspectives regarding animals in scientific research. Because Indigenous communities lie at the nexus of complex legal, political, and cultural intersections, we must draw on several sets of literature to inform this discussion and its implications for animal sciences. This question requires some engagement with philosophy of science and related questions raised by Native American scholars.

We review three areas of literature relating to Indigenous perspectives in animal sciences. First, we explore the human–animal bond, looking at animal interventions in therapy and education and human–dog relationships in Indigenous communities. Second, we look at genetic and breeding programs that affect and are in partnership with Indigenous communities. Third, we review areas of Traditional Ecological Knowledge (TEK) that focus on studies relevant to animal sciences. Several questions emerge from this literature review: Why are there so few Indigenous perspectives published in animal sciences? And how does this literature review, in conversation with Native studies, help to explain this gap? What is the potential for including Indigenous sciences in animal sciences?

Who Is Indigenous?

Before defining Indigenous/Native science, we must understand the term Indigenous. In this piece, we define the word Indigenous using the United Nations (1) definition:

> Indigenous peoples are inheritors and practitioners of unique cultures and ways of relating to people and the environment. They have retained social, cultural, economic and political characteristics that are distinct from those of the dominant societies in which they live. Despite their cultural differences, indigenous peoples from around the world share common problems related to the protection of their rights as distinct peoples.

Indigenous persons are also place-based peoples with distinct, precolonial societal structures, cultures, and languages, along with historic connections to a particular land base. In the United States, Native American tribes fall under the United Nations's category of Indigenous persons. Because of this, we use Native and Indigenous interchangeably to discuss Indigenous communities in North America. There are 574 federally recognized Native American tribes in the United States. When a tribe is specified, we refer to that Indigenous community by name. Lastly, for Australian or Canadian Indigenous communities, the terms Aboriginal or First Nation might be used in place of Native or Indigenous.

Indigenous/Native Science

Indigenous/Native science is an emerging term developed in Native American studies and Native studies–adjacent fields and has grown in pockets among a diverse set of disciplines. Other disciplines like education, law, literature, and philosophy have adopted "Indigenous" or "Native" in front of their discipline to explain distinctly Native and Indigenous thought within those

particular fields. Native or Indigenous knowledges have also been applied to more technical disciplines like ecology, forestry, and animal-assisted therapy. Most commonly, these studies and research articles fall under the category of TEK, but not always. There has been an acceleration of literature about TEK emerging in environmental sciences; however, the authors identify a significant gap in TEK that does not focus on Indigenous perspectives and values in animal research.

The discussion of knowledge, research, and science cannot be understood apart from the influence of colonization within Native North American communities. The discussion of colonization is important for understanding why scholars describe an inherent difference of worldview between Indigenous and non-Indigenous communities. One of the most important texts to address the foundations of this philosophy in research and education is Smith's (2) book *Decolonizing Methodologies: Research and Indigenous Peoples*, in which she argues that there are fundamental differences between Indigenous and academic communities' production of knowledge. Smith explains how academic research about Native people has been used to characterize Native communities as "primitive" societies. Such characterizations have justified the erasure of Native knowledge systems and replaced them with European-influenced or Western ways of knowing. American Indian scholars in education have published widely on this tension (2–6). For more specific examples of this tension and its implication for animal-focused sciences, John (7), Weisiger (8), and Hubbard (9) have written about the US government's forced slaughter of North American animal communities and the destruction this brought to Native community cultures.

In an article focused on science education, Snively & Corsigilia (10) argue that teaching Western science as "universal" is problematic and argue for a more nuanced critique on the standardized methodology of scientific inquiry. They write, "Westerners freely acknowledge the existence of indigenous art, music, literature, drama, and political and economic systems in indigenous cultures, but somehow fail to apprehend and appreciate indigenous science" (10, p. 7). We borrow from their definition of "Indigenous/Native science" to start our discussion. Indigenous science, they write, "interprets how the world works from a particular cultural perspective" (10, p. 8). They also note that scientific processes are "woven into all aspects of Indigenous cultures" (10, p. 8).

Cajete (11) wrote one of the foundational books addressing the intersection of Indigenous epistemologies and scientific inquiry: *Native Science: Natural Laws of Interdependence*. He emphasizes the differences between the methods of acquiring knowledge in Western systems verses Native systems that focus on relationality and interconnected paradigms of inquiry. The introduction of specifically Indigenous or Native sciences is situated within the discussion of Native epistemologies or meaning-making systems. One of the driving forces of colonialism was to discredit Native ways of interacting with land and animals as a justification for forced removals of peoples and lands.

Indigenous science is made up of a set of agreements shared between tribal communities; we call these values, which serve as the guiding principles for the implementation of research and education with and for Indigenous peoples. These values common to most, if not all, tribal epistemologies have been summarized in many different publications; we cite a few important texts here to outline four core principles of Indigenous sciences and their references in the literature.

The first core principle is the belief that Native worlds (human, plant, and animal) are deeply interconnected. There is an inherent relationship between land, animal, and human communities within Native worldviews. Cajete (11, p. 64) calls this "dynamic multidimensional harmony." This was explained to the authors, who are Navajo, in the Navajo context: When we make an adjustment to one entity or being, that adjustment does not happen in isolation but has a ripple effect on all other beings. Pierotti & Wildcat (12) explain this interconnectedness as it pertains to TEK. They write, "One major theme of TEK is that all things are connected, which is not simply a homily or a romanticized cliche, but instead is a realization that no single organism can exist without the

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web of other life forms that surround it and make its existence possible" (12, p. 1336). The theme of interconnectedness is similar to current conversations about One Health paradigms currently popular in agriculture, veterinary medicine, and many other areas of research. Chalmers & Dell (13) make the connection between One Health and research with tribal epistemologies in an article on One Health and animal interventions.

The second shared principle is a rejection of the human/animal hierarchy inherent in many Western disciplines (14), meaning that humans do not have dominion over the management and control of animal beings. To further explain this hierarchy, scholars have referenced philosophical influences from Western philosophers including John Locke and René Descartes (15). Roncancio et al. (14) examine textbooks and critique the ontological foundation of human–animal separation and its impact on animal husbandry education. They write, "The textbook examining animal husbandry was concerned only with animals as a form of wealth creation, i.e., live 'stock', reducing their existence to units of exchange. Little regard was placed upon engagement between and among the more-than-human and human world or even animal welfare" (14, p. 8). In the Native worldview, animals and plants are persons who have agency, autonomy, and an ability to understand humans and the world around them (15, 16).

The third core principle is that Native sciences are grounded in the values of the community, which may have inherent obligations to relationships and protocols that are not accepted in Western scientific methods. There is a spiritual element to Indigenous human–animal relationships that has been improperly represented in Western research:

> Relationships with animals have always been an important part of Native American spirituality. These relationships (often called the "worship" of animals) are considered a "primitive" stage of human intellectual development by Western science. But the core idea is being rediscovered through the environmental movement or eco-philosophy; that is the idea that we are all related, that humans have a responsibility to animals. (11, p. 168)

Spiritual practices of Indigenous persons not only are religious practices but are embodied by the lived experience of science and spirituality combined. So, the songs, stories, prayers, and ceremonies required for animal interactions are both spiritual and scientific at the same time. This also means that non-Native researchers working with Native communities are responsible for navigating those community protocols. This is known as tribal consultation, whereby a community might offer the respectful protocols that are necessary to engage in such research. Tribal consultation can take years before any research is ever started.

Lastly, the fourth value is an understanding that animals in Indigenous communities are family or kin relatives (12). Cajete (11, p. 159) describes an intimacy between humans and animals, writing that "these understandings formed the basis for an ecological ethic of such a depth and intimacy that it continues to have profound impact on contemporary Indigenous people." That intimacy means there is a basis of accountability. One is accountable to a deer in the woods in the same way one is accountable to a human family member. This intimacy and shared kinship also creates the foundation for a mutually beneficial relationship between humans and animals. Pierotti & Wildcat (12, p. 1334) call this the "third alternative," where there is a give-and-take between humans and nature; they frame it as both extractive and conservationist.

What Indigenous Science Is Not

Along with describing what Native/Indigenous science is, it is also important to describe what Native science is not. Self-identifying Native people who train and conduct research in various scientific communities, like animal sciences, does not necessarily mean these researchers are doing Native/Indigenous science. This is because their work might not be in partnership with

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Native communities or center Native epistemologies. Several noteworthy intersections are important, meaning that Native or non-Native researchers working with Indigenous communities walk the line of Native/Indigenous science insofar as they are navigating the needs, ethics, relationships, and values of that community. Navigating these areas often results in the need to adjust methodology to fit the needs of the community; these adjustments, conversations, and strategic translations/relationship building are part of the Native science process but do not stand alone.

Who Has Access to or Owns Indigenous Science?

Indigenous science is grounded in the specific interrelated epistemology articulated by precolonial tribal societies. One must be in close, meaningful relation to Indigenous communities to engage with Indigenous knowledges. This is usually, but not always, Indigenous community members. Indigenous scientific knowledge is legally owned by the tribal community and exists as communally oriented knowledge. It is not owned by an individual researcher, university, or research team, and many tribes have protocols in place to protect their knowledges, so they are owned by the tribal nation. When these protocols are unclear, the researcher is responsible for consulting the tribe about the ownership and publication of knowledge and data. This article reviews several projects in which Indigenous and non-Indigenous persons partnered together for research.

Indigenous Science and Animal Science

In our search for Indigenous perspectives in animal sciences, we looked (at the most basic level) for articles written about Indigenous societies, and even more specifically for research that centered an Indigenous point of view. This work is best done when researchers have close and meaningful relationships to Native communities and allow for the work to center the tribal community's relational system of knowledge. Indigenous science differs from Indigenous studies because most methods in Indigenous studies draw from social sciences and humanities and tend to center questions about peoples and lands rather than animals. This creates a significant gap in Indigenous studies literature that does not center animals. Further addressing this gap, Schneider (17, p. 151), who writes about human–fish relations, argues that not enough attention has been paid to political land relationships in animal sciences:

> Too little attention has been paid to the specificity of place and how settler colonialism shapes the lives and deaths of the plants and animals of that place—and can even shape the place itself. I argue that the root of both issues is the assumption arising from European views of property, or land, as generic: in other words, the tendency to treat one acre of forest as equivalent to any other acre of forest when organizing space.

Not only is there a gap in Indigenous perspectives represented in animal sciences, but there is a significant gap in Indigenous studies around animal research and studies in partnership with animal scientists. Locating research that confronts social, political, and relational networks of humans/land/animal beings within Indigenous communities, while centering Indigenous values, is a challenge.

Search Terms and Methods for Searching

We conducted a search in 10 of the leading animal science journals. The following search terms were used to search titles, abstracts, and full texts: "tribal/tribe," "Native American," "American Indian," and "Traditional Ecological Knowledge." We did not use the search term "Indigenous" because of the different meaning in biological literature. Based on our search terms, no articles were found in the following journals: *Animal Bioscience*, *Journal of Veterinary Medicine*, *Journal of*

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Animal Ecology, and *Mammal Review*. *Animal* mentioned TEK, but in relation to African Indigenous communities and not Native North American communities. Several articles were found in the *Annual Review of Animal Biosciences* and *Agriculture Ecosystems and the Environment*, but those only mentioned Native Americans in passing, with no significant engagement with the contemporary Native community or knowledge systems. Only three articles from the *Journal of Animal Science* and the *Journal of Agriculture, Food Systems, and Community Development* mention Native American or American Indian tribes in their titles or abstracts. These titles include "Breeding for Adaptability to Local Conditions with Special Reference to the Navajo Indian Reservation" (18); "Genetic Diversity in and Conservation Strategy Considerations for Navajo Churro Sheep" (19); and "Between Drought and Disparity: American Indian Farmers, Resource Bureaucracy, and Climate Vulnerability in the Southern Plains" (20). These articles are included in the following review of literature.

For a broader search, several combinations of the following terms/phrases were used in Google Scholar and the University of Arizona library search engine: "animal science" and "animals" in combination with "tribal/tribe," "Native American," "American Indian," and "Traditional Ecological Knowledge." We read abstracts and searched articles to determine whether the article was about American Indian communities or merely mentioned "American Indian" or "Native American" in passing without pertaining to these populations. From this search, we found a much higher number of articles in a selection of journals, including *Science Education*, *Ecological Applications*, *EcoHealth*, *Anthrozoös*, *Sustainability*, *American Indian Culture and Research Journal*, *Human Ecology*, *Pimatisiwin: A Journal of Indigenous and Aboriginal Community Health*, and *The Journal of Wildlife Management*.

Although the search was not exhaustive, it illustrates the difficulty in finding research articles that address Native perspectives in animal sciences and suggests a large gap in the literature regarding Native community participation and perspectives in animal sciences. Given the lack of publications in top animal science journals, we include several publications in a diverse set of journals that highlight Native perspectives as they relate to animal sciences. We also include a few commentary articles that address these tensions. We classified the literature on Indigenous perspectives we did find into three areas of animal sciences: the human–animal bond, breeds and genetic testing, and TEK.

HUMAN–ANIMAL BOND

We examined literature regarding the human–animal bond and found articles about animal-assisted interventions in learning and therapy along with studies focused on dog–human relations. Most articles fall under the umbrella of human–animal relations and focus on companion species: equines and dogs. We believe this is because of the ongoing legacy of both the horse and the dog in many Native societies, as well as the explosion of research on human relations with horses and dogs in human–animal bonds.

In an earlier article that addresses Native perspectives on the human–animal bond, Constable et al. (21) conducted interviews with Indigenous humans in Australia to document relations and attitudes among humans and dogs. Although this research took place in Australia, we include this study because the philosophical foundations that inform North American TEK and Indigenous science are heavily informed by the literature written by Indigenous communities in Australia. The authors argue that there is a complex nexus of relations between Aboriginal persons and their dog companions. They situate these relations within the backdrop of the community's cultural orientation as well as the legacy of colonialism for land, humans, and animal persons. The article confronts the assumption that Indigenous humans do not care about their dog companions due

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to the high number of free-roaming dogs on Indigenous lands. They explain that a forced change in relationships to land and the introduction of settler agriculture have significantly impacted Indigenous human–dog relations in the present moment. The study found that most humans and communities recognized a significant bond with dogs, describing them as family and community members. Although communities recognized the free-range and health issues, "the necessity of culturally appropriate solutions to these issues was emphasized" (21, p. 344). This article emphasizes the importance of culturally appropriate solutions for existing animal problems on tribal lands.

In another study, Aenishaenslin et al. (22) used a mixed-methods approach to understand human–dog relations and the implications for health and wellness. The authors express concerns about free-roaming dogs in remote Inuit communities and the potential for disease transmission. The study interviewed and surveyed both Inuit and non-Inuit people, finding that Inuit people expressed the cultural significance of dogs more than non-Inuit respondents. The study also suggests that the lack of access to veterinary services contributes to a difference in dog-owning practices between Inuit and non-Inuit people. The practice of free-roaming dogs is a culturally grounded method of domestication for Inuit people, and the authors argue that culture should be taken into consideration for "decision making when implementing interventions to mitigate dog-related health risks" (22, p. 158).

Chalmers and Dell describe the use of a One Health paradigm as a theoretical framework that has aided their work with Indigenous communities in animal-assisted interventions. They argue that animal-assisted interventions pay insufficient attention to environment and focus solely on animal interactions. They suggest a One Health framework might be useful for the future of animal-assisted interventions, while paying attention to the alignments that One Health paradigms have with Indigenous community worldviews.

Two articles demonstrate the development of Indigenous value–driven equine-assisted learning (EAL). Dell et al. (23) documented the development of an EAL program grounded in the First Nation's culturally relative methods. The research team worked with the White Buffalo Youth Inhalant Treatment Centre, which serves First Nations youth recovering from substance abuse. While creating the EAL program, the authors (23) describe the importance of health as "holistic" and "culturally-based," to promote resiliency for Indigenous communities impacted by colonialism. Because of the horse's significance in many Indigenous communities, "the horse externally validates traditional values the youth learn in treatment programming; the horse puts into practice cultural teachings (e.g., horse reacts uniquely to youth who are expressing frustration)" (23, p. 95). This program illustrates points of connection between First Nation and Western methods of health and recovery, while also being careful to center the specific importance of culturally based methods for First Nations. The authors suggest next steps, one of the most important being a need for "empirical evidence surrounding the effectiveness of animal-assisted interventions, and in particular EAL" (23, p. 98), especially for methods used in partnership with First Nation's culturally based models of health.

Snowshoe & Starblanket (24) developed a tribally specific mental health framework for equine-facilitated therapy. They begin by describing the importance of culture for First Nation mental health practices due to the ongoing legacy and violence of colonialism. They also agree with Dell et al.'s assertion that far too few studies consider the effectiveness of equine-assisted interventions for First Nations communities. Interestingly, the authors include a section in their article titled "Decolonizing the History of the Horse," in which they argue that the horse was an original companion to Indigenous communities and not introduced by the Spanish. This is an important element to the cultural relationship of Indigenous communities and horses because it illustrates the importance of horses in creation, ceremony, and science. As mentioned in the Introduction,

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the combination of spiritual and scientific knowledge makes up the entirety of Indigenous sciences. In Snowshoe & Starblanket's (24) methods, they "decolonise equine assisted approaches" by centering the horse as a "teacher and healer" instead of a tool used to intervene in human health. Researchers use a blanket metaphor to describe four areas of importance: taking a trauma-informed perspective, using a strengths-based approach, meeting community-specific needs, and being spiritually grounded. Ultimately, the article links the survival of Lac La Croix Indigenous ponies to the survival of First Nations humans.

Finally, Martorelli et al. (25) studied a pilot program called Sunka Wakan, which incorporated equine activities into a holistic program for student well-being. The program centered the cultural significance of the horse in Lakota/Nakota/Dakota tribal nations and introduced the equine as a partner. The program included five phases. The first two focused on the introduction of Native American horse–human relationships and "cocreative language building" with equine activities. The third phase included activities that "supported students who were beginning to process their experiences in school by providing students with events that affirmed their presence at the university, while creating a respectful environment for them to learn and grow" (25, p. 751). Phases three and four focused on healing and processing. Phase five helped students prepare for their transition with a trail ride on which they partnered with a horse for the "celebration of their journey together throughout the year and to symbolize the continuation of their journey of self-awareness and care through their college experience" (25, p. 751).

All of these studies' assisted interventions included culturally grounded paradigms and work with scholars in the fields of education and psychology as well as community members. Many of the studies used mixed-methods approaches to ground the importance of cultural and spiritual animal relations within Indigenous perspectives. We believe they center and include Indigenous perspectives on the human–animal bond, representing the community in meaningful and respectful ways through programming, research, and publication. However, the majority of these studies exist as one-time or pilot programs, and a significant gap remains in culturally based animal-assisted interventions and culturally based human–animal bond research.

BREEDS AND GENETIC TESTING

There is some information on the effects of breeding and genetic testing as it relates to Indigenous communities and their values. Some of the articles found are empirical studies, whereas others are projects in process or anecdotal evidence. Many of these projects do not go into detail about the cultural or spiritual significance of animals within the Native communities they work with or represent.

"Genetic Diversity in and Conservation Strategy Considerations for Navajo Churro Sheep" is a study conducted by Maiwashe & Blackburn (19) to investigate the status of the genetic diversity for the Navajo-Churro sheep breed, which has implications for the Navajo Nation because of the animal's cultural and economic importance. Genetic diversity in sheep is the result of sexual reproduction that gives rise to offspring by combining genes from parents. These offspring give rise to different physical attributes for the individual and the capacity to adapt to stress, diseases, and unfavorable environmental conditions. One outcome of the study showed that small herds are more likely to have higher incidence of inbreeding, thereby reducing genetic diversity compared to larger herds. Navajo sheep ranchers typically have smaller herds, most notably due to the Livestock Reduction program in the 1930s. Prior to the Livestock Reduction program, Navajos maintained large sheep herds, and those herds likely maintained excellent herd health due to increased genetic diversity. Today, Navajos keep individual animals for shorter periods of time and introduce new animals to the herd to reduce inbreeding. Historical trauma due to the Livestock

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Reduction program has impacted Navajo sheep ranchers, as they had to adapt to a new way of raising their prized Churro sheep.

Stroupe et al.'s (26) "Genomic Evaluation of Hybridization in Historic and Modern North American Bison (*Bison bison*)" highlighted the bison's importance to Native American peoples. Bison numbered in the millions in North America and Canada until targeted killing by settlers (9), loss of access to suitable habitat, and death by epidemic caused by exposure to imported and native infectious diseases dramatically reduced their numbers. Fortunately, some ranches captured wild calves and established herds. Unfortunately, the practice of hybridizing bison with various domestic cattle breeds compromised the integrity of bison as a species. Genetic testing of bison involving mitochondrial DNA sequencing, as well as nuclear microsatellite markers, showed evidence of matrilineal cattle ancestry. However, some bison herds were found to be free from cattle introgression. Stroupe et al.'s study used whole-genome sequencing to better assess the shared derived alleles between cattle and bison. The results showed that bison identified as "free of cattle genes" actually contain some cattle genes. The study has increased efforts to find a "true" precontact bison species. Based on the history of tribal breed programs like the programs mentioned below, we believe there is interest among tribes in understanding original breeds of the Americas that exist without colonial intervention.

Moulton (27) discusses Lewis and Clark's documentation indicating that Nez Perce Appaloosa horses were abundant in the Columbia River Plateau in the Pacific Northwest area and the explorers' claim that "their horses appear to be of an excellent race; they are lofty elegantly formed active and durable." Later, local ranchers sought out these horses due to their quality. The conflict with the US government and eventual removal of the Nez Perce from their land led to the elimination of the Appaloosa. In 1994, the Nez Perce tribe in Idaho started a horse breeding program with the idea of crossbreeding Appaloosa horses with a Central Asian breed called Akhal-Teke to produce a new breed, the Nez Perce Horse. The horse breeding program is located in Lapwai, Idaho, and seeks to restore the traditional horse breeding culture of the Nez Perce. The Nez Perce horse breeding program was jointly financed by the Nez Perce tribe, the US Department of Health and Human Services, and a nonprofit called the First Nations Development Institute. The Nez Perce National Historical Park has archives of collections, books, and other documents about the horse history of the Nez Perce. This history demonstrates the interspecies violence of colonialism, and the rich knowledge the tribe possessed to breed a quality horse.

Although there are no scientific publications, a news article shared Navajo Nation mustang genetic testing results found by partners from the Navajo Nation and Texas A&M researchers. The *Navajo-Hopi Observer* (28) published an article sharing some of the results of the genetic testing on Navajo mustangs. The reporter described how Navajo Nation Division of Natural Resources executive director "Shebala provided an extensive history of horses on the Navajo Nation and described how understanding their genetic identities will help integrate more durable long-term solutions to the current problems." The article describes how the process of breed identification through genetic testing is a potential pathway for the Navajo Nation to implement solutions to a long-standing high feral horse population.

All of these studies are conducted by non-Native researchers with some (or no) relation to Native communities, and although they do not represent Indigenous or Native science per se, they include meaningful tensions faced in Indigenous communities. In fact, because there are very few examples of Indigenous-grounded animal genetic studies, researchers must proceed with tribal consultation and adhere to protocols recommended by tribes. We urge researchers to develop more operationalized methods of data storage and management to partner with tribes and protect tribal data sovereignty.

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TRADITIONAL ECOLOGICAL KNOWLEDGE AND ANIMALS

Traditional Ecological Knowledge (TEK) is the most popular key phrase to find Indigenous knowledge integrated into scientific literature. Pierotti & Wildcat (12, p. 1335) explain four main points of thought shared among tribal nations in the development of TEK:

> This way of thought includes: (1) respect for nonhuman entities as individuals, (2) the existence of bonds between humans and nonhumans, including incorporation of nonhumans into ethical codes of behavior, (3) the importance of local places, and (4) the recognition of humans as part of the ecological system, rather than as separate from and defining the existence of that system.

Snively & Corsigilia (10, p. 21) have a few helpful observations about the development of TEK as a "branch of biological and ecological science that has become known as the TEK, which can be thought of as either the knowledge itself, or as documented ethno-science enriched with analysis and explication provided by natural science specialists." They recognize the many definitions and practices that fall under the umbrella of TEK and its inherent connection to critiques of Western-based science.

In TEK literature, only a handful of articles address animal–human relations within tribal communities. We reviewed a couple of articles that center the agency and personhood of animals in relation to place and humans. However, we found that most TEK literature is land based. Most of the TEK articles that mention animals describe them in relational proximity to place or land or describe a general importance of animals in the interconnected paradigm of TEK. We presume this to be because so few TEK studies exist in the animal sciences, so there are not many pathways for collaboration between animal science professionals and TEK knowledge communities.

Much of the TEK literature addressing animals focuses on hunting practices within Indigenous communities. One example is an article titled "Hunting and Morality as Elements of Traditional Ecological Knowledge," published in *Human Ecology* (29). The authors interviewed hunters to understand how moral values from Indigenous communities inform their contemporary hunting practice. Their goal is to "illuminate the relationships between practical and ethical dimensions of TEK" (29, p. 16).

Polfus et al. (30) conducted a comparative study examining the effectiveness of TEK and Western science methods to "spatially predict the occurrence of collared caribou locations," finding that TEK was a valuable methodology. They argue that TEK has implications for wildlife management. Our hope is that TEK literature will push beyond simply proving that TEK is an effective method compared with Western science and begin to showcase the value of TEK while centering Indigenous communities, values, needs, and epistemologies in the development of research. Polfus et al. argue that TEK has implications for wildlife management. Bang et al. (31, p. 150) take up a similar critique of TEK in their article titled "If Indigenous People Stand with the Sciences, Will Scientists Stand with Us?" They argue that TEK knowledge is usually taken seriously only when "it has been 'verified' by modern science." Instead, their hope, and ours, is that TEK and Indigenous science can stand on its own and in meaningful partnership with Western sciences.

Lastly, in their work in TEK, Bhattacharyya & Slocombe (32) describe equine agency–centering Indigenous perspectives through an Indigenous worldview. This article differs from other TEK articles because it focuses on the agency of horses in what the authors call a "kin-centric ecology." They stressed the importance of animal agency for scientific study and the opportunity for animals to make decisions in scientific processes. They use qualitative methods to conduct a case study with the Xeni Gwet'in and Yunesit'in First Nation communities in British Colombia, Canada. The community describes the ongoing relationship to wild horses and their animal husbandry. Bhattacharyya & Slocombe (32, p. 9) summarize,

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> One major point that emerged from this study was that uneven power relations in management are not only about who gets to make decisions over human use of land and resources, but also about the role that the land itself, and the animals that live on it, plays in those decisions, and how they will be represented.

Besides a selection of studies outlined here, animals are mostly just mentioned in TEK literature and not typically central to the studies. Additionally, TEK studies that partner with animal scientists to express more than a comparative affirmation of TEK information are rare.

CONCLUSION

The studies above are a selection of Indigenous perspectives in animal sciences, but not exhaustive. Most literature on Indigenous perspectives about animals still exists in the fields of social science, philosophy, Indigenous and American Indian studies, and education. Aside from the aforementioned studies, most TEK articles do not directly partner with animal sciences. Articles in animal science journals showcase minimal Indigenous community involvement and do not ground any studies in Indigenous or Native science. The one exception to this seems to be literature in human–animal bonds, where the cultural and spiritual significance of animals is taken seriously and represented mostly through qualitative methods. Similarly, genetic studies that involve Native communities do not center the work within the values and tensions of that community and could benefit from an engagement with "data sovereignty" (33). We agree with Snively & Corsigilia (10, p. 8), who argue,

> there are a number of issues that make it difficult to incorporate indigenous science examples into a Western scientific framework. Chief among them is the fact that many scientists and science educators continue to view the contributions of indigenous science as "useful," but outside the realm of "real science."

They believe the reason for this is twofold: first, the belief that science is a universal rather than a relative concept and, second, Western science's discomfort with spiritual components of Indigenous science (10).

The most robust engagement with Indigenous sciences at present comes from environmental sciences, human–animal bonds, and ecology. We believe this to be because the inclusion and engagement of Indigenous knowledges are understood as relational in nature and require disciplines that are familiar with trans-disciplinarity and relational knowledge as a source. Pierotti & Wildcat (12, p. 1336) provide a similar analysis, arguing that TEK and ecology tend to share some similar foundational principles of interconnectedness. Given the interconnected paradigm of Indigenous worldviews, we suggest that the ongoing siloes of scientific disciplines might be a contributor to the dearth of Indigenous perspectives. We also believe that the One Health paradigm is a potential starting point to connect with Indigenous communities around interconnected paradigms. However, much work remains to be done to center Indigenous perspectives on the meaning of interconnectedness.

We also believe the lack of literature in animal sciences could be due, in part, to the lack of Native American people within the animal sciences. There is a need to continually increase the presence of Native scholars entering and pursuing careers in animal sciences. More curriculum, research, and literature that center Indigenous science might support the retention of Indigenous students in these fields. We also encourage the investment of diversity support and training for colleges that house animal sciences.

Suggestions

Animal scientists should consider next steps necessary to include and center Indigenous perspectives within animal science fields. Given that human–animal relations are foundational within

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Native communities, these communities and their worldviews must be respected in research and education. Furthermore, Indigenous communities have vast amounts of animal husbandry and relational knowledge that they have crafted over many thousands of years in relation to animal persons and place.

Fundamental to the inclusion of Indigenous and Native scientific perspectives are good relations with tribal communities. We thus conclude with several suggestions.

First, scientists interested in working with Indigenous communities must adhere to the legally binding and required tribal consultation policies put in place by each federally recognized tribal nation within the United States and Canada. These processes are sometimes communicated by institutions through formal Tribal Consultation Policies. A good example of a university policy can be found at the University of Arizona on the Native American Advancements, Initiatives, and Research page (34). For more on tribal consultation, we recommend reading "Guidelines for Considering Traditional Knowledges (TKs) in Climate Change Initiatives," which summarizes best practices for working with tribes and implementing traditional and Indigenous sciences (35).

Second, we encourage animal scientists to be aware of the political context in which tribal nations operate. They are distinct sovereign nations that have unique and complicated histories with the US government, Department of the Interior, and institutions of higher education. We recommend that researchers in animal sciences include social scientists with expertise in tribal relations before beginning work with tribal communities.

Lastly, we recommend more scholarly spaces to continue conversations about Indigenous perspectives in animal sciences. A special issue journal or a symposium focused on Indigenous perspectives in animal sciences would be helpful to build networks of scholars doing similar work in pockets among US institutions and tribal nations. Creating academic places to start these conversations will help Indigenous sciences gain more visibility and respect in animal sciences. We imagine this to be an interdisciplinary space in which social and physical sciences can communicate across disciplinary boundaries.

DISCLOSURE STATEMENT

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LITERATURE CITED

1. United Nations. 2022. *Indigenous peoples at the United Nations*. Resour., Dep. Econ. Soc. Aff., United Nations, New York. **https://www.un.org/development/desa/indigenouspeoples/about-us.html**
2. Smith LT. 2012. *Decolonizing Methodologies: Research and Indigenous Peoples*. London: Zed Books
3. Brayboy BMJ. 2021. Tribal critical race theory: an origin story and future directions. In *Handbook of Critical Race Theory in Education*, ed. M Lynn, AD Dixson, pp. 191–202. New York: Routledge
4. Smith LT, Tuck E, Yang KW, eds. 2018. *Indigenous and Decolonizing Studies in Education*. New York: Routledge
5. Tachine AR. 2017. Grandmothers' pedagogy: lessons for supporting Native students' attendance at universities. In *Indigenous Pathways, Transitions and Participation in Higher Education*, ed. J Frawley, S Larkin, JA Smith, pp. 151–67. Singapore: Springer
6. Sabzalian L. 2019. *Indigenous Children's Survivance in Public Schools*. New York: Routledge
7. John KD. 2019. Animal colonialism: illustrating intersections between animal studies and settler colonial studies through Diné horsemanship. *Humanimalia* 19(2):42–68
8. Weisiger M. 2011. *Dreaming of Sheep in Navajo Country*. Seattle: Univ. Wash. Press
9. Hubbard N. 2014. Buffalo genocide in nineteenth century North America: "kill, skin, and sell." In *Colonial Genocide in Indigenous North America*, ed. A Woolford, J Benvenuto, AL Hinton, pp. 292–305. Durham, NC: Duke Univ. Press

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10. Snively G, Corsigilia J. 2001. Discovering indigenous science: implications for science education. *Sci. Educ.* 85:6–34
11. Cajete G. 2000. *Native Science: Natural Laws of Interdependence*. Santa Fe, NM: Clear Light Publ.
12. Pierotti R, Wildcat D. 2000. Traditional ecological knowledge: the third alternative (commentary). *Ecol. Appl.* 10(5):1330–40
13. Chalmers D, Dell CA. 2015. Applying One Health to the study of animal assisted interventions. *EcoHealth* 12:560–62
14. Roncancio IV, Temper L, Sterlin J, Smolyar NL, Sellers S, et al. 2019. From the Anthropocene to mutual thriving: an agenda for higher education in the Ecozoic. *Sustainability* 11(12):3312
15. John KD. 2022. What does it mean when Indigenous peoples say animals are sacred? In *Indigenous Religious Traditions in Five Minutes*, ed. M Basset, N Avalos, pp. 146–48. Sheffield, UK: Equinox Publ.
16. Fienup-Riordan A. 2020. *Uqlautekevkenaku*: They didn't make a mess of it: Yup'ik perspectives on human and animal relations. In *They Say They Have Ears Through the Ground: Animal Essays from Southwest Alaska*, pp. 1–26. Fairbanks: Univ. Alaska Press
17. Schneider L. 2013. There's something in the water: salmon runs and settler colonialism in the Columbia River. *Am. Indian Cult. Res. J.* 37(2):149–64
18. Cooper JM. 1939. Breeding for adaptability to local conditions with special reference to the Navajo Indian Reservation. *J. Anim. Sci.* 1939(1):26–32
19. Maiwashe AN, Blackburn HD. 2004. Genetic diversity in and conservation strategy considerations for Navajo Churro sheep. *J. Anim. Sci.* 82(10):2900–6
20. VanWinkle TN, Friedman J. 2019. Between drought and disparity: American Indian farmers, resource bureaucracy, and climate vulnerability in the Southern Plains. *J. Agric. Food Syst. Commun. Dev.* 9(B):53–68
21. Constable S, Dixon R, Dixon R. 2010. For the love of dog: the human dog in rural and remote Australian Indigenous communities. *Anthrozoös* 23(4):337–49
22. Aenishaenslin C, Brunet P, Leévesque F, Gouin GG, Saint-Charles ASJ, et al. 2019. Understanding the connections between dogs, health, and Inuit through a mixed methods study. *EcoHealth* 16:151–60
23. Dell CA, Chalmers D, Dell D, Suave E, MacKinnen T. 2008. Horse as healer: examination of equine assisted learning in the healing of First Nations youth from solvent abuse. *Pimatisiwin* 6(1):81–106
24. Snowshoe A, Starblanket NV. 2016. Eyininiw mistatimwak: the role of the Lac La Croix Indigenous pony for First Nations youth mental wellness. *J. Indig. Wellbeing* 1(2):6
25. Martorelli CB, Eastman A, Oyate (Dakota) S-W, Mook N, Bott-Knutson RC. 2018. Sunka Wakan: the horse as a partner in welcoming American Indian students to the campus community. *J. Coll. Stud. Dev.* 59(6):749–52
26. Stroupe S, Forgacs D, Harris A, Derr JN, Davis BW. 2022. Genomic evaluation of hybridization in historic and modern North American bison (*Bison bison*). *Sci. Rep.* 12:6397
27. Moulton GE. 2018. *The Lewis and Clark Expedition: Day by Day*. Lincoln: Univ. Neb. Press
28. *Navajo-Hopi Observer*. 2020. Navajo horse program identifies 32 genetically diverse horse breeds. *Navajo Hopi Observer*, July 20. **https://www.nhonews.com/news/2020/jun/30/navajo-horse-program-identifies-32-genetically-div/**
29. Reo NJ, Whyte KP. 2011. Hunting and morality as elements of traditional ecological knowledge. *Hum. Ecol.* 40:15–27
30. Polfus JL, Heinemeyer K, Hebblewhite M, Taku River Tlingsit First Nation. 2014. Comparing traditional ecological knowledge and western science woodland caribou habitat models. *J. Wildl. Manag.* 78(1):112–21
31. Bang M, Marin A, Medin D. 2018. If Indigenous people stand with the sciences, will scientists stand with us? *Daedalus* 147(2):148–59
32. Bhattacharyya J, Slocombe S. 2017. Animal agency: wildlife management from a kincentric perspective. *Ecosphere* 8(10):e01978
33. Kukutai T, Taylor J. 2016. *Indigenous Data Sovereignty: Toward an Agenda*. Canberra, Aust.: ANU Press
34. Ariz. Board Regents. 2016. *ABOR 1-118 Trial Consultation Policy*. **https://naair.arizona.edu/research-engagement-guidelines**
35. Clim. Tradit. Knowl. Workgr. 2014. *Guidelines for considering traditional knowledges (TKs) in climate change initiatives*. Guidel., Clim. Tradit. Knowl. Workgr., Sept. **https://climatetkw.wordpress.com/**

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